

Introduction

- Impaired autophagy, a cellular process in which **damaged organelles are recycled**, is associated with many **degenerative diseases and disorders** (ex: Alzheimer's and Parkinson's Disease)(Glick, Barth, & Macleod, 2010).
- These diseases affect over 55 million people worldwide.
- As an individual ages, the **autophagy process decreases**.

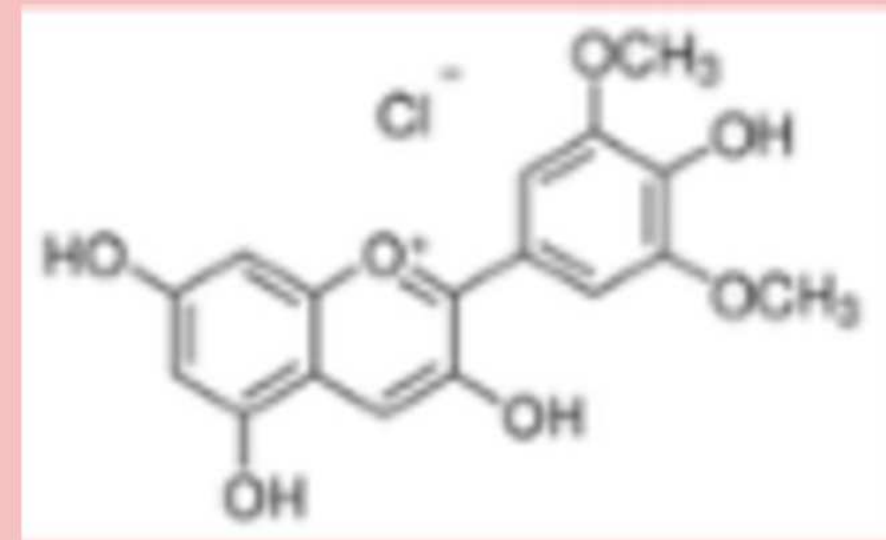


Figure 1. Chemical composition of Malvidin Chloride

- Polyphenols, compounds that provide the color for many fruits and vegetables, have had **numerous benefits on lifespan**, though the cause is not entirely certain.
- One such polyphenol in blueberries, anthocyanin (specifically named **Malvidin Chloride**), was observed to determine its effect on the autophagy process at various concentrations.
- This polyphenol has shown **antioxidative, anticancer, and antimicrobial** properties previously, making it a viable option to increase cellular health through autophagy (Meccariello & D'Angelo, 2021).

Purpose

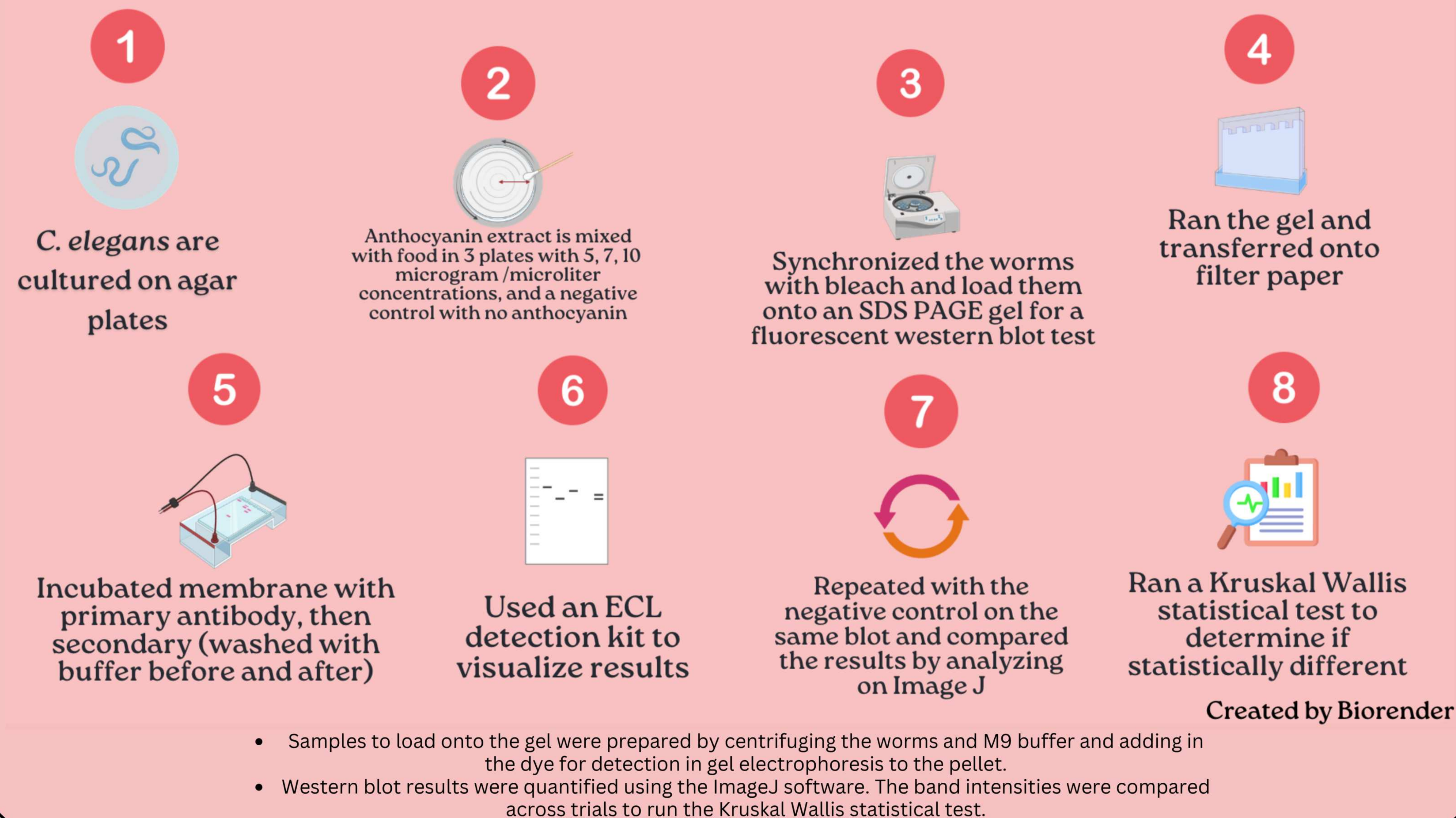
- The connection between polyphenols and autophagy is not entirely unknown, especially as concentrations increase, so this experiment was conducted to determine if the consumption of anthocyanin by *C. elegans* (nematodes that share **2/3 the same genes** as humans) would result in an increase in the autophagy process in the nematodes.
- This was measured through observing the **LGG-1 protein** since an abundance of the protein is associated with increased autophagy.
- It was hypothesized that as the concentration of Malvidin Chloride (**IV**) administered to *C. elegans* increases, the **LGG-1 protein will be expressed more (DV)**, indicating an increase in autophagy.



Figure 2. Image of *C. elegans* under 400X magnification

Effect of Varying Concentrations of Blueberry Anthocyanin Extract, Malvidin Chloride, on Mitochondrial Health in *C. elegans*

Materials and Methods



Results

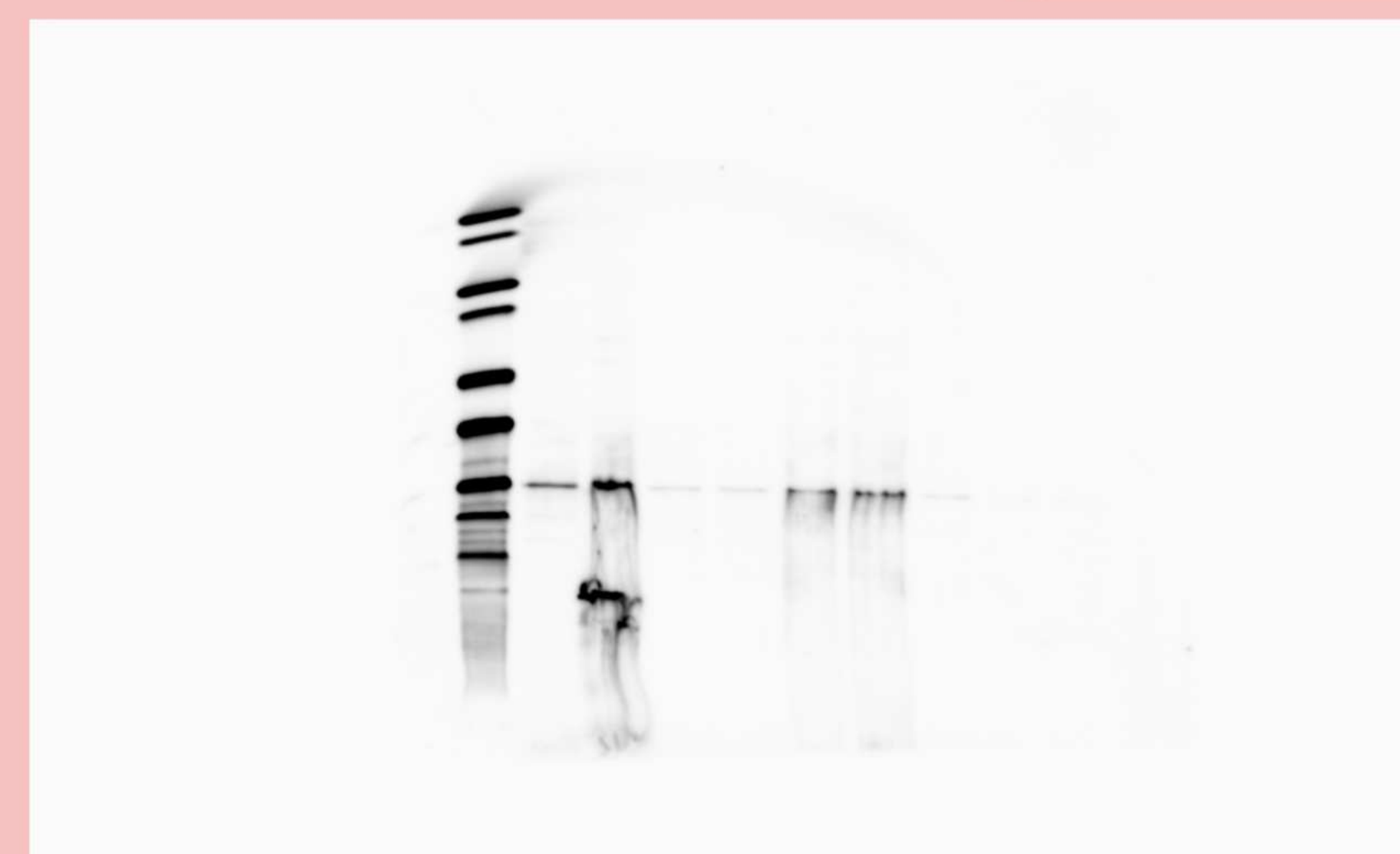


Figure 3. Image of the western blot for trial 1.

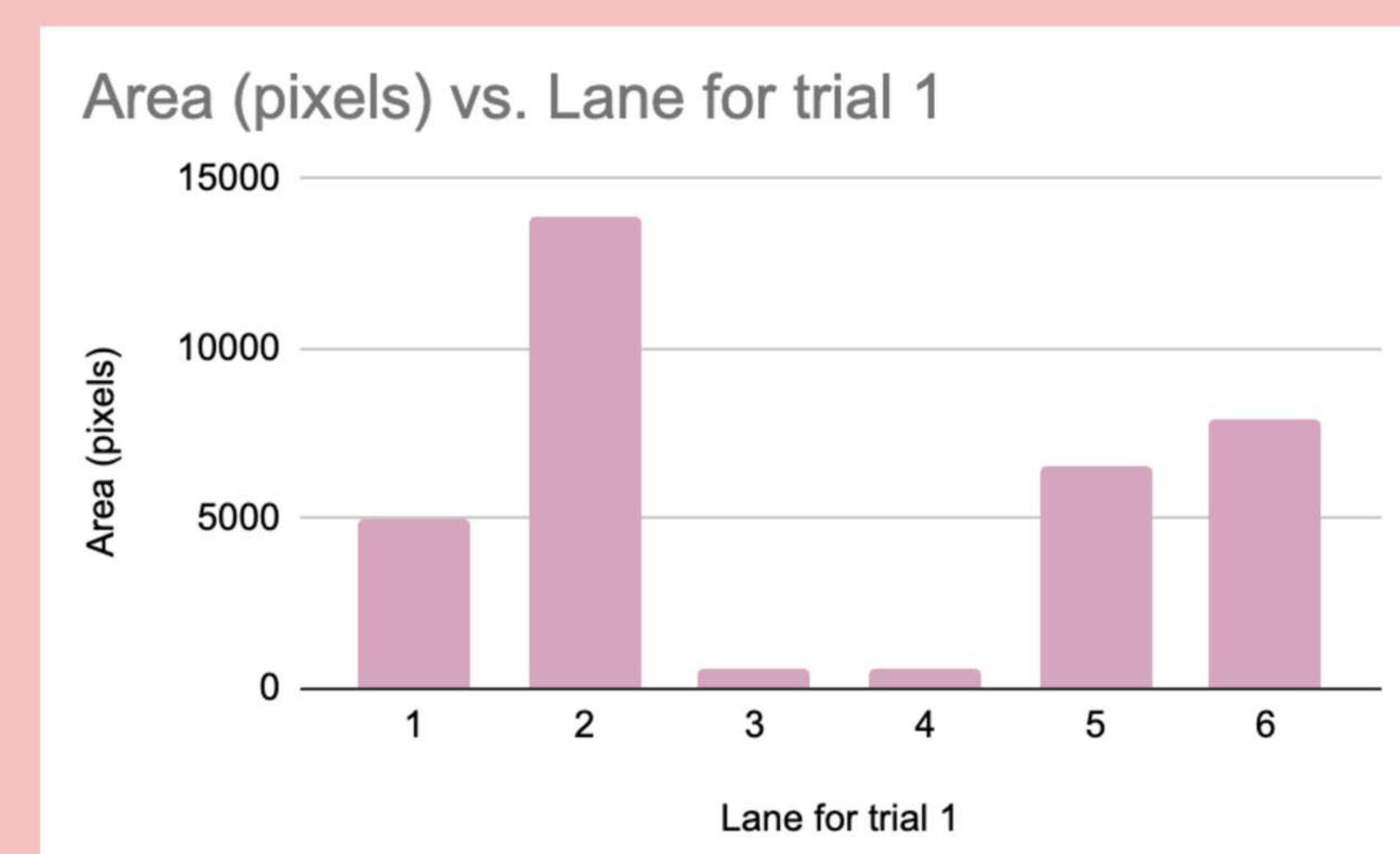


Figure 5. Graph of the pixel area per lane in trial 1. Lanes 1 and 2 are the negative control, lanes 3-5 are a concentration of 5 µg/µL of Malvidin Chloride, and lane 6 is a concentration of 7 µg/µL.

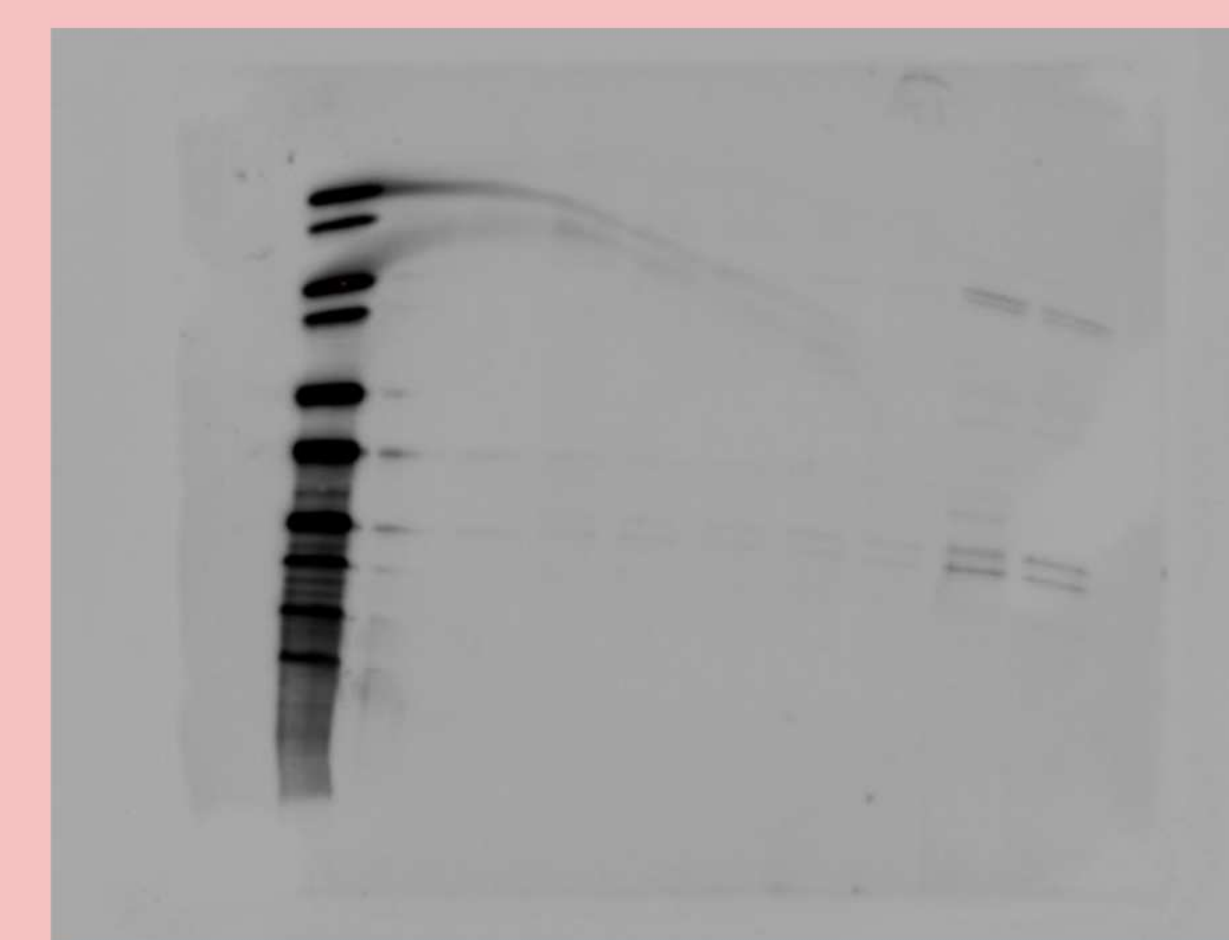


Figure 4. Image of the western blot for trial 2.



Figure 6. Graph of the pixel area per lane in trial 2. Lanes 1 and 2 are a concentration of 10 µg/µL of Malvidin Chloride, lanes 3 and 5 are a concentration of 7 µg/µL, lanes 4 and 6 are a concentration of 5 µg/µL, and lanes 7-9 are the negative control.

Analysis and Conclusions

A **Kruskal-Wallis test** was conducted to **compare the percent pixel area** among the **5, 7, and 10 µg/µL concentration groups**. The test showed **no statistically significant difference** between concentrations ($H = 5.66, p = 0.059$), indicating that concentration did not significantly affect pixel percentage under these conditions. This indicates that there is **no significant difference in the LGG-1 protein production** of the nematodes between the different concentration exposures. This could be due to the fact that there were only 2 clear trials, and the bands were not very visible in both. Additionally, the samples created did not have many nematodes since the plates the samples were made from didn't have an abundance of healthy worms.

Further Research

- Further research could focus on testing **different forms** and types of anthocyanin to determine how the LGG-1 protein is expressed differently
- Additionally, nematodes at different life stages could be tested to determine at which stage they are most affected.

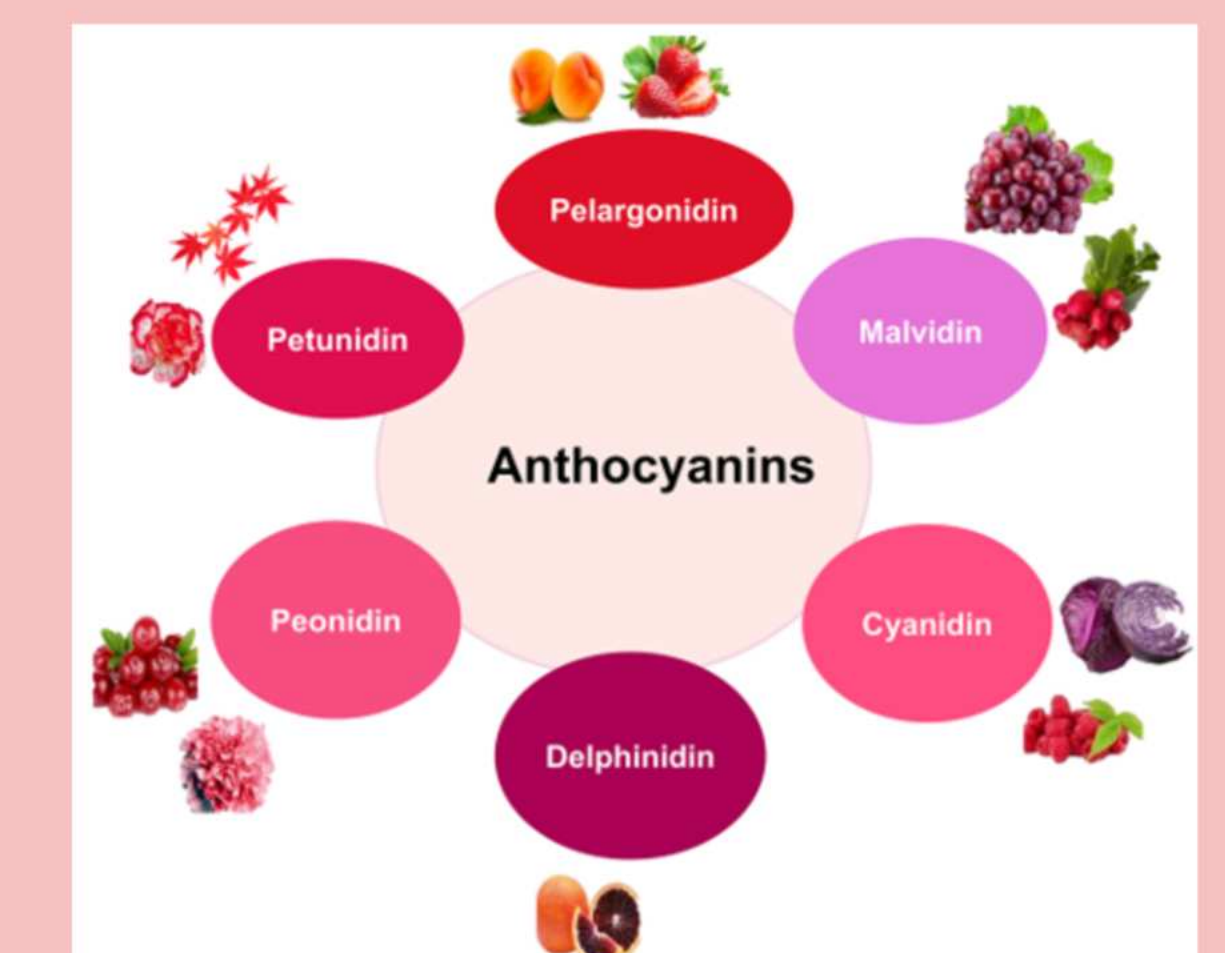


Figure 7. Image of the other types of anthocyanins present in fruit

References

- Glick, D., Barth, S., & Macleod, K. F. (2010). Autophagy: Cellular and Molecular Mechanisms. *The Journal of Pathology*, 221(1), 3–12. doi:10.1002/path.2697
- Meccariello, R., & D'Angelo, S. (2021). Impact of polyphenolic-food on longevity: An elixir of life. *Antioxidants*, 10(4), 507. doi:10.3390/antiox10040507
- Palmisano, N. J., & Meléndez, A. (2016). Detection of autophagy in *Caenorhabditis elegans* by western blotting analysis of LGG-1. *Cold Spring Harbor Protocols*, 2016(2). <https://doi.org/10.1101/pdb.prot086512>
- Zhang, S. (2012, January 1). Cell isolation and culture. *WormBook: The Online Review of C. elegans Biology* [Internet]. <https://www.ncbi.nlm.nih.gov/books/NBK153594/>